

Mapping IT Skills Trends: An Analysis of LinkedIn Job Listings

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Abstract: The rapid evolution of the Information Technology (IT) sector demands continuous adaptation from professionals and organizations, yet labor market intelligence often relies on fragmented and unstructured data. This study addresses the challenge of systematically extracting and standardizing job posting information to identify skill demand and its relationship with salary outcomes. Focusing on LinkedIn job listings for IT roles, we developed an automated framework that combines web scraping and Large Language Models (LLMs) with prompt engineering to extract and normalize skills and compensation data. Our results reveal strong correlations between specialized skills—particularly in machine learning, cloud computing, and DevOps—and higher median salaries, while also highlighting the persistent relevance of foundational programming and database skills. The contributions of this work are twofold: academically, it advances labor market analysis by introducing a scalable, domain-agnostic methodology for structuring unstructured labor data; practically, it provides actionable insights to support individual career planning and the design of organizational training programs.

Keywords: LLM; IT skills; labor market; LinkedIn; job postings; IT salary

1. Introduction

The rapid evolution of the Information Technology (IT) sector has created a dynamic job market that demands continuous adaptation from individuals and organizations [1–3]. Knowledge is central to global economic transformation and organizational success, reshaping how it is produced, disseminated, and consumed amid the rise of advanced services and immaterial labor [4–6]. Recognizing the value of measuring intellectual assets underscores the growing importance of organizational knowledge as a competitive asset [7,8]. Effective Knowledge Management (KM) is critical for enhancing decision-making, increasing efficiency, and fostering innovation. For IT employees, refining essential technical skills through career management and continuing education remains key to staying competitive [9].

While prior NLP and text-mining studies have shown that elements of job profiles and postings can be structured, they typically do not jointly extract and normalize heterogeneous skill mentions and compensation information—often expressed in diverse units and periods—from open-ended descriptions at scale. Therefore, the central research problem of this study is: *How can unstructured labor market data from LinkedIn job postings be effectively extracted and standardized—particularly skills and salary expressions—to identify skill demand and its relationship with salary outcomes in the IT sector?*

This work analyzes labor market data through job postings from LinkedIn for various IT roles, a professional networking platform widely used for job searching and recruiting

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[10]. The methodology combines automated web scraping and the use of Large Language Models (LLMs) [11] to extract key skills and salary information from these postings, enabling the identification of skills most strongly correlated with higher salaries. This approach supports the development of a comprehensive dataset highlighting in-demand IT job market skills and offers guidance for career planning and organizational training programs. Prior LinkedIn analyses have mainly relied on text processing or topic extraction to categorize postings and surface in-demand skills. This study differs by jointly extracting and standardizing both skill mentions and salary expressions from unstructured postings via an LLM-based pipeline, normalizing heterogeneous compensation units and periods to enable integrated analysis. Building on this standardized dataset, we move beyond pairwise associations to skill clustering: (i) co-occurrence clusters that reveal how competencies are bundled in job ads, and (ii) salary-based clusters that stratify skills by their associated median pay. [12,13]

Therefore, the contributions of this study are twofold. On the academic side, it advances labor market analysis by developing an LLM-based approach, supported by prompt engineering, to extract and standardize unstructured data from LinkedIn job postings. This methodological framework addresses the lack of structured information on the platform and is designed to be flexible and domain-agnostic, enabling scalable skill and compensation mapping across sectors. On the practical side, it generates actionable insights into IT job market trends by identifying the most in-demand technical skills and their associated salary ranges, providing guidance for both career planning and the design of targeted training programs.

By clearly addressing the methodological gap in labor market analytics and delivering actionable insights for practice, this study positions itself at the intersection of knowledge management, artificial intelligence, and labor economics.

2. Related Works

Developing technical skills and fostering a culture of continuous learning are essential for growth in the IT sector. Ajayi and Udeh [14] employ a systematic literature review to investigate workforce upskilling initiatives for emerging technologies in IT. The study recognizes the constantly changing nature of the industry and highlights the importance of adopting strategic, inclusive, and flexible training approaches to effectively bridge the skills gap and ensure sustainable growth in the IT sector. Furthermore, research by Falck et al. [15] shows that IT skills yield strong economic returns, particularly when aligned with market demands, underscoring the need for targeted skill development strategies.

IT skills are increasingly vital across different fields beyond traditional tech sectors. In healthcare, Altawaty et al. [16] emphasize the need for tailored technology training programs. Their study identifies key training priorities through expert opinions and statistical analysis, including recognizing e-health as a specialty and equipping workers with hands-on experience in relevant medical technologies. Their work highlighted the need for context-specific training initiatives that address challenges shaped by industry and geographical location, emphasizing the necessity of adaptable approaches to skill development. Pastrana and Tob [17] conducted a socioformative analysis of leading IT companies, highlighting the importance of aligning training programs with emerging technological trends. Their analysis shows the need to understand industry-specific needs and incorporate socio-cultural factors into training designs as strategic factors for fostering a skilled workforce capable of driving innovation in the knowledge society.

Eggenberger and Backes-Gellne [18] explore the impact of IT skills on labor market outcomes following involuntary job separations. Their study distinguishes between generic and expert IT skills, finding that while generic IT skills enhance worker adaptability and

reduce earnings losses, expert IT skills may exacerbate earnings declines, especially for workers with highly specific skill bundles. The authors utilized occupational training curricula and Natural Language Processing (NLP) combined with employment data to analyze the long-term effects of job separations on earnings and career trajectories. Using econometric modeling and accounting for factors like education, industry, and prior work experience, the authors highlight how different IT skills shape career paths. Their findings reveal the varied impact of IT skills on career stability, suggesting that effective skill development must consider both market-wide trends and the unique demands of specific job roles.

Lyu and Liu [19] used online job postings to investigate the balance between soft and hard skills in the labor market, particularly within the energy sector. While there has been a noticeable trend toward emphasizing soft skills, their analysis, which involved data from 2010 to 2019, suggests that prioritizing hard skills may be more beneficial for productivity, particularly in roles requiring technical expertise. The Organisation for Economic Co-operation and Development (OECD) also used job postings to analyze skill demands in the labor market. Their findings indicate a rise in specific IT skills, particularly those related to AI [20]. Employer demand for graduate skills across four states in the United States is also examined, focusing on IT roles, revealing that employers are increasingly open to hiring graduates from non-IT fields if they possess relevant technical skills [21]. These results highlight the critical role of universities in Knowledge Management and the need to align educational programs with evolving industry demands to better equip graduates for the job market.

Lyu and Liu [19] use online job postings in the energy sector (2010–2019) to examine the balance between soft and hard skills and find that hard skills remain closely tied to productivity in technical roles. The OECD analysis [20] likewise relies on vacancy data to document rising demand for specific IT skills, particularly AI-related ones. Evidence on employer demand for graduates across four U.S. states further suggests openness to hiring non-IT graduates who possess relevant technical competencies [21]. These studies primarily measure skill incidence or employer stated preferences; they do not link positions and skills to compensation.

Several studies in the literature have examined LinkedIn data to uncover hiring patterns, skill demands, and market dynamics, reinforcing the platform's relevance in labor market research. Earlier approaches, however, mainly rely on traditional NLP techniques focused on classification or profile matching rather than extracting structured compensation and skill data. López-Carril et al. [22] highlight LinkedIn's role in professional development, showing how platform use can bridge academic learning and industry expectations. Bastian et al. [13] apply topic extraction to infer skills and competencies from user profiles, enabling career-path suggestions. Li et al. [23] build upon this by applying deep learning to standardize job postings into structured fields (e.g., job titles, skills, company), improving LinkedIn's job-matching algorithms. These works demonstrate the value of structuring LinkedIn data but remain centered on matching people to jobs rather than analyzing which skills drive compensation.

Kazi et al. [12] also use traditional text processing and machine learning (Naïve Bayes and SVM) to classify job descriptions into categories such as Android, iOS, and web development. Their study shows the feasibility of extracting signals from unstructured postings, yet focuses only on categorization and not on linking skills to salary outcomes. The study is capable of analyzing profiles and categorize postings. However, they do not link extracted skills to job context or compensation. This gap motivates approaches capable of structuring unstructured postings into analyzable elements. Our study leverages LLMs

to automatically extract both skills and compensation data, enabling a more integrated analysis of job requirements and salary structures.

3. Methodology

The methodology employed in this work involved automated web scraping of LinkedIn job postings for various IT roles. We gathered job posting data from LinkedIn, specifically targeting IT roles. The data includes work type (hybrid, on-site, remote), job level (associate, entry, mid/senior), posting's title, company, location, posting link, job description, and the searched position and date of the search. Though internship, director, and executive levels are options, they were not considered in this work. The employer provides the job description, but lacks a standardized format. Employers may use it to outline role challenges, qualifications, skills, experience, and certifications needed. It also details job responsibilities, work environment, team dynamics, and career growth opportunities, helping candidates assess their suitability and interest.

To clarify the data collection process, all job postings were obtained from LinkedIn on June 22, 2024, using thirteen predefined IT-related job titles as search strings (e.g., Data Scientist, DevOps Engineer, Software Engineer). Each search combined the chosen job title with LinkedIn's filters for work type (hybrid, on-site, remote) and job level (entry, associate, mid/senior), restricted to postings from the past month. For every retrieved posting, we stored the title, company, location, job description, posting link, work type, job level, the corresponding search string, and date. Duplicates were removed using job title, company, and level as a unique key. This conservative key may occasionally create false duplicates, but we prioritized avoiding overcounting. Internship, director, and executive roles were also excluded to reduce noise and ensure comparability of skill and salary requirements across technical positions.

Although LinkedIn was chosen as the sole data source in this study, its global reach, structured filtering options, and high volume of IT-related job postings ensure both relevance and scalability. Nevertheless, relying exclusively on LinkedIn constitutes a limitation, as it may not capture regional or sector-specific opportunities advertised on other platforms. Future work could extend the dataset by incorporating postings from additional sources such as Indeed, Glassdoor, or specialized IT job boards to provide a broader and more diverse view of the labor market.

The data was collected using an automated web scraping approach. As all search queries were IT-related, the resulting data sets are not mutually exclusive. Duplicates were removed based on the job posting's title, company name, and job level. Therefore, the number of results for each query may not be completely accurate. Automated duplicate removal may bias results across queries, leading to potential discrepancies in the data.

Using Google's Gemini [24], we extracted relevant skills and salary information, when available, from the job descriptions through the Application Programming Interface. Listing 1 shows the prompt used for the skills extraction. Although other extraction techniques could yield different results, their impact in this context is expected to be minimal. This is because each block of information processed is small and highly specific, ensuring that the LLM's context is not overloaded. As a result, the prompt design becomes a more decisive factor than the choice of model, and any potential bias from the LLM is unlikely to affect the outcomes significantly. More advanced models could be more efficient, but the focus here is on demonstrating data processing capability rather than benchmarking performance.

The purpose of the first prompt is to request a highly specific task while also allowing for flexibility, enabling adjustments to avoid redundancies. This approach ensures that typos and extraneous characters that do not belong to the skill name can be removed. Listing 2 shows the prompt used for the salary extraction.

Since the data extraction is automated, the prompt requires the output to be specifically formatted. It must also account for how salaries are conveyed across different job descriptions. The expected output is a list containing each job's salary period, minimum, and maximum range. If only the minimum range is provided (e.g., "salary of 100k+", "from 50k", etc.), the maximum range is assumed to be the same as the minimum. Conversely, if only the maximum range is provided, it is considered that no salary information is given. To enable the analysis of this work, all values are asked to be converted to a numeric format. All salary figures are analyzed in nominal terms as posted, without purchasing-power or cost-of-living harmonization across locations; while a purchasing power parity adjustment would strengthen cross-regional comparability, it was beyond the scope of this exploratory study.

Even though the prompt requested that the skills be standardized, not all issues were resolved. A list of all skills identified by Gemini was extracted and stored in a text file. This file was input into OpenAI's GPT-4o model. ChatGPT was tasked with re-standardizing the skills and returning a new text file to update the list of extracted skills. Listing 3 shows the prompt used.

The midpoint of the posted salary range, computed as $(\text{minimum} + \text{maximum})/2$, was used as the reference salary. Finally, all the data used in this work was available in a tabular format through the Python scripts, the Gemini API, and the web scraping algorithm.

However, though most cases had the skill name correct, some errors persisted. However, we presented the results based on the 50 most frequent skills. With this filter applied, there were no redundancy issues. Our method's code and materials are available at Zenodo [25]. No manual audit of the LinkedIn extraction (precision and recall) and no confidence intervals or significance tests were included, as these fall outside our descriptive, exploratory scope.

K-Means was selected for the clustering tasks because it provides a straightforward and interpretable way of grouping skills in terms of co-occurrence and salary association. Alternative approaches like DBSCAN or agglomerative clustering can capture irregular cluster shapes or hierarchical structures. However, they typically require more sensitive parameter tuning and may produce less balanced groups in high-dimensional tabular data. By contrast, K-Means offers a balance of simplicity, scalability, and transparency, making it suitable for this exploratory setting. The specific outcomes of applying K-Means are discussed in detail in the next section.

We acknowledge that the methodology employed in this work does not constitute a methodological innovation, as it combines established web scraping procedures and prompt-based information extraction with large language models. However, we also note that methodological novelty is not always required for a study to be valid and valuable. In this case, the aim was to design an appropriate and reproducible process capable of transforming unstructured job descriptions into analyzable data. This approach could be

```
USER: Imagine you're an NER AI model. Your task is to extract technical
      skills, frameworks, languages, software, and concepts from the given job
      posting.
Be concise and focus only on the relevant skills required for the candidate.
Do not include outcomes or results of using these skills.
Standardize the names of skills and software to ensure clarity and
      meaningfulness. You can remove or add text to do so.
Provide a single list of required skills.
List 1-3 word entities only, without sentences. Format your response as a
      list.
USER: Here is the description: JOB DESCRIPTION
```

Listing 1. Gemini prompt for skill extraction.

USER: Check if the salary range is mentioned in the job description. If it is mentioned and already specified per year, return [yearly, minimum range, maximum range].

If the rate is hourly, convert it to yearly by multiplying by 2080 and return [hourly, minimum range, maximum range], with the ranges converted to yearly values.

If the rate is monthly, convert it to yearly by multiplying by 12 and return [monthly, minimum range, maximum range], with the ranges converted to yearly values.

If only the minimum range is provided, repeat the value for the maximum range. If only the maximum range is provided, consider the salary as not provided and return 'not specified'.

Consider different terms, such as salary, compensation, pay, in the context of the description. Convert minimum range and maximum range to numbers. For instance, if the value is \$10k, convert it to 10000.

If more than one range is provided, return the first one. Include a salary range even if the compensation value is location-specific. If no salary range is specified, return exactly 'not specified'.

USER: Here is the description: **JOB DESCRIPTION**

Listing 2. Gemini prompt for salary extraction.

USER: I have a list of skills in a text file that I need to standardize. Can you help me with removing any special characters from each skill, normalizing capitalization so that each skill is consistently formatted, standardizing terms that mean the same thing (e.g., ‘JavaScript’ and ‘javascript’ should both be ‘JavaScript’), handling acronyms properly by ensuring they are not repeated (e.g., ‘VPN’ and ‘virtual private network’ should not both be on the list), removing duplicates after standardizing the skills, and providing the cleaned and standardized list in the same order as a text file. The list of skills is in the attached text file. Please make sure to be thorough in your restandardization process.

Listing 3. ChatGPT prompt for skill standardization.

expanded for future research by systematically comparing different LLMs, integrating hybrid pipelines that combine LLMs with traditional NLP techniques, or applying clustering and topic-modeling methods to reveal broader patterns in the demand for skills.

4. Results

Table 1 (left) summarizes the distribution of IT-related job titles—used as search strings—by employer-specified experience level. At the time of collection (a snapshot of LinkedIn postings), business intelligence analyst was the most common title (731 listings), while blockchain developer was the least common (268). Because job titles are tightly coupled with required skills, this skew in postings—and any residual bias from automated deduplication—likely tilts the skill mix toward certain groups (e.g., data/analytics) and should be kept in mind when interpreting the results.

Table 1 (right) reports the same breakdown but only for postings where the LLM successfully extracted salary information. In this subset, business intelligence analyst remains the most frequent, whereas machine learning engineer appears fewer than 100 times. The dominant experience band is mid/senior level, consistent with the right-skewed salary distribution in Figure 1. The snapshot nature of the data and the imbalance in job titles result in an over-representation of skills tied to the most frequent roles.

Table 1. Position Data: all listings (left) and listings with salary (right).

Position	Associate	Entry Level	Mid Senior	Total	Associate	Entry Level	Mid Senior	Total
Blockchain Developer	44	40	184	268	21	15	92	128
BI Analyst	206	237	288	731	38	68	76	182
Cloud Solutions Architect	110	176	339	625	32	35	72	139
Cyber Security Specialist	154	176	209	539	32	34	48	114
Data Scientist	158	194	141	493	48	66	42	156
Database Administrator	152	119	323	594	47	38	67	152
DevOps Engineer	160	183	218	561	59	50	66	175
IT Support Specialist	106	176	149	431	31	49	49	129
ML Engineer	74	117	78	269	30	45	24	99
Network Administrator	176	212	222	610	53	49	44	146
Software Developer	117	75	189	381	31	27	73	131
Systems Analyst	103	357	262	722	13	90	66	169
Web Developer	164	195	156	515	47	63	27	137
Total	1724	2257	2758	6739	482	629	746	1857

Figure 1 presents the histogram of the reference salaries distribution. The distribution depicted in the plot is notably right-skewed, indicating that most salary observations are clustered toward the lower end of the salary range, but fewer observations extend into the higher salary ranges. Most salary data points are concentrated between \$50,000 and \$150,000, with the highest frequency observed around the \$100,000 mark.

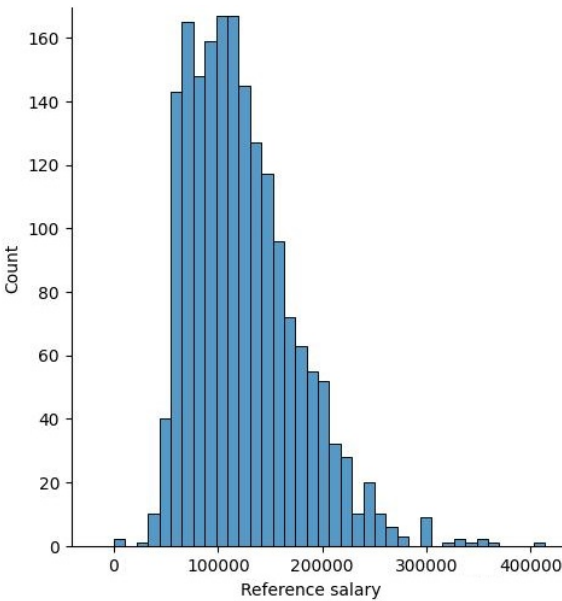


Figure 1. Salary distribution (outliers omitted).

Figure 2 illustrates the salary distribution across the different job levels. Mid/senior level positions exhibit the highest salary range, with more than 75% of listings under this category having the midpoint of salary range above \$100,000. Therefore, we can see a clear relation exists between job level and salary.

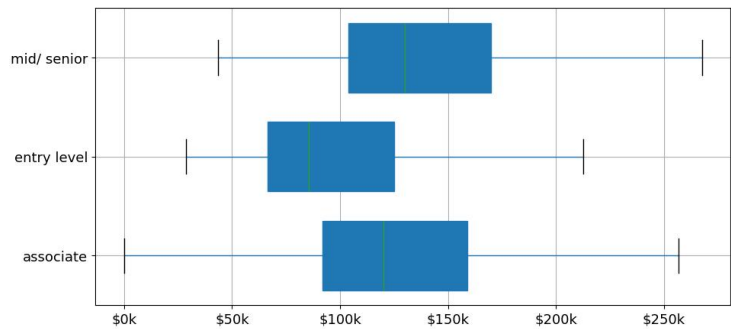


Figure 2. Box plot of salaries for each job level.

Figure 3 presents a box plot highlighting the wide range of salaries across the different positions, used as search strings. Roles associated with emerging or highly specialized domains—such as blockchain developer, machine learning engineer, and cloud solutions architect—exhibit higher medians and broader salary ranges. This pattern aligns with labor-market evidence that rapid technological change generates skill premiums for workers able to adapt to evolving technologies [26]. Additionally, firms operating in innovation-intensive environments tend to pay higher wages to attract and retain workers with scarce technical skills, especially in areas where innovation is a strategic priority [27].

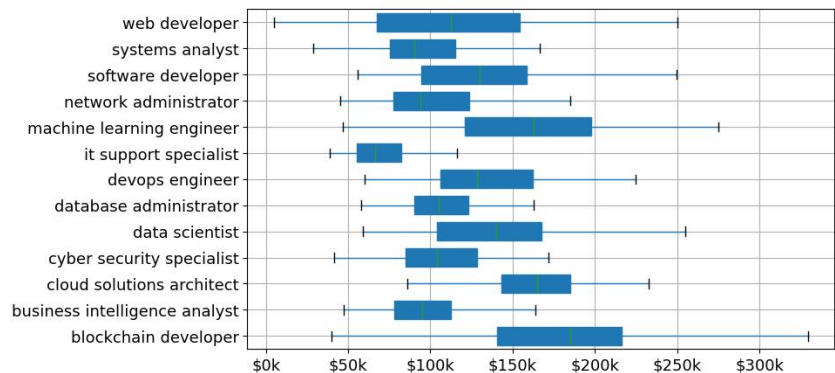


Figure 3. Box plot of salaries for each position (search query).

Figure 4 displays the bar chart of the top 50 skills, ranked by their frequency of occurrence. At the top of the chart, Python stands out with the highest frequency of 1,735 occurrences. Following Python, SQL has a significant count of 1,565, reflecting its essential role in database management and data manipulation. Amazon Web Services (AWS) comes in third with 1,188 occurrences. Beyond reflecting popularity, these skills indicate a stratified labor market centered on data-centric and cloud-oriented roles. LinkedIn postings at the time of data collection consistently favored skills related to data-driven decision making and scalable system deployment. This is expected, since Table 1 shows that positions requiring these skills—such as Business Intelligence Analyst and Cloud Solutions Architect—are among the most frequently posted in the dataset.

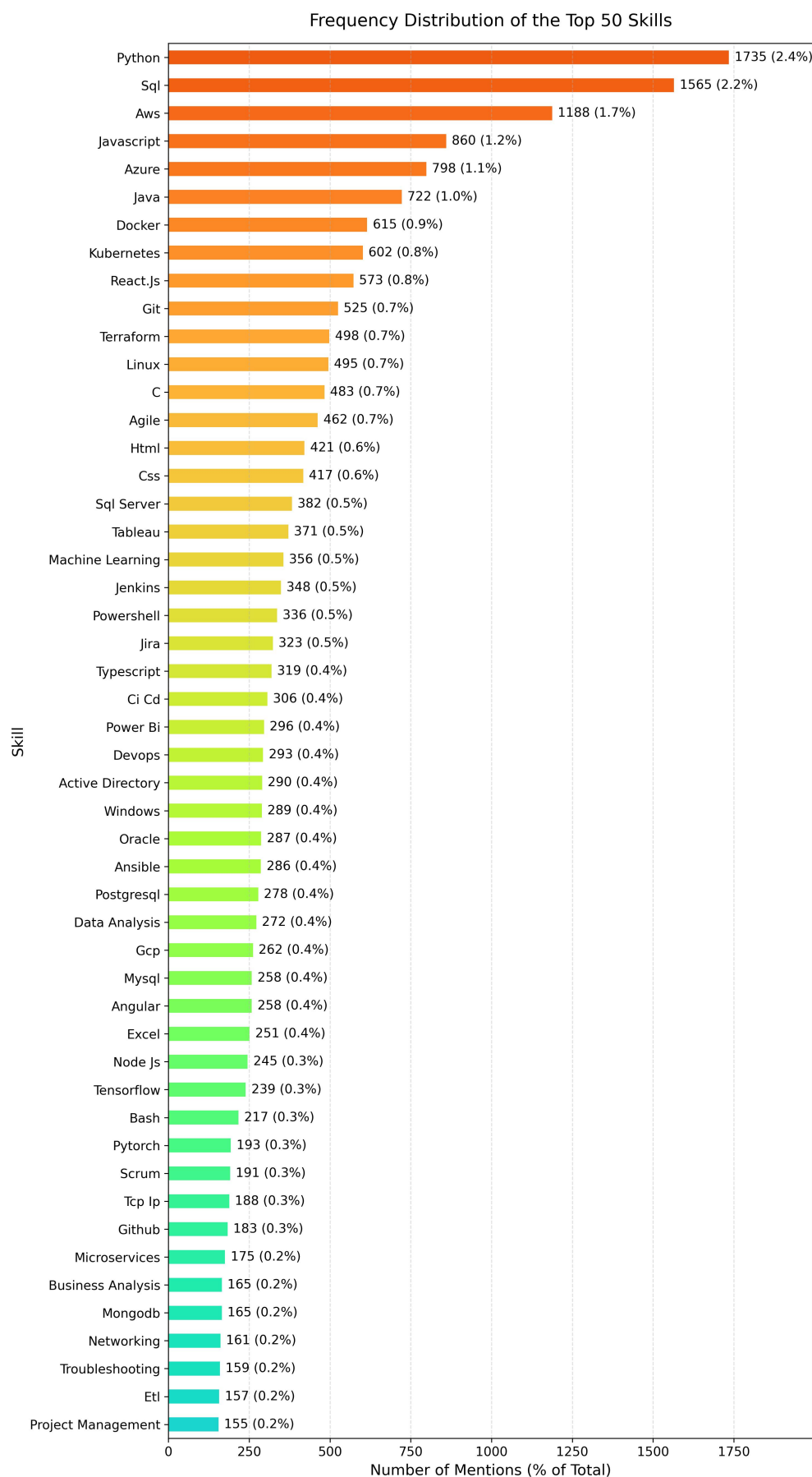


Figure 4. Frequency plot of the Top 50 skills.

Traditional skills remain highly relevant in the job market. JavaScript (860 occurrences), Java (722), and C (483) continue to form the backbone of software development, showing their enduring importance in building and maintaining systems. Likewise, HTML (421), CSS (417), and SQL Server (382) are essential for web development and database management. Rather than being replaced by emerging AI and big-data technologies, these foundational skills sustain an entire segment of the labor market. Their prevalence in job postings suggests the coexistence of different role profiles.

Figure 5 presents the salary distribution box plot for the top 50 skills, highlighting the significant variation in compensation associated with different skills in the tech industry. Skills such as AWS, TensorFlow, and PyTorch are associated with higher median salaries, indicating high demand and value in the job market. As relatively new skills, there may be a shortage of qualified professionals, driving up their market value. In contrast, while still essential, more traditional skills like Windows, SQL, and HTML tend to show narrower salary ranges and lower medians. This data underscores the dynamic nature of tech salaries, where expertise in cutting-edge technologies can lead to significantly higher earnings.

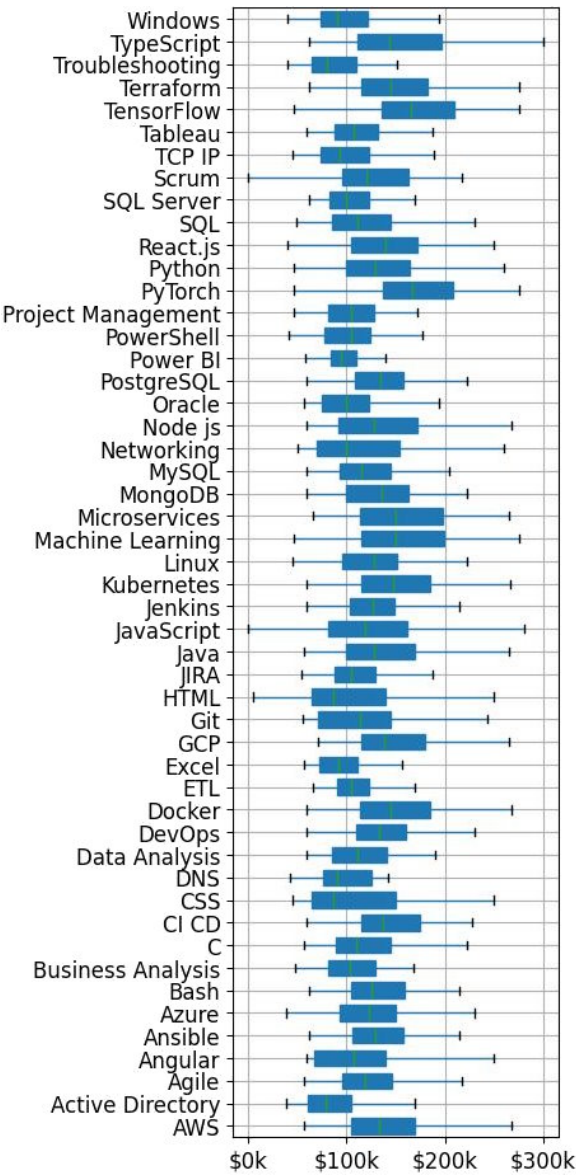


Figure 5. Box plot of salaries of the Top 50 skills.

To explore the relationship between individual skills and salary, we computed the point-biserial correlation [28]. This metric is a special case of the Pearson correlation designed for situations where one variable is binary (e.g., whether a job posting requires a specific skill) and the other is continuous (salary). While it is suitable for assessing the association between the presence of a skill and salary, this approach assumes a linear relationship and evaluates each skill independently. In practice, many skills tend to co-occur—such as Python, TensorFlow, and PyTorch—so the apparent effect of one skill may be partially explained by another. Therefore, point-biserial correlations do not capture interaction effects or multicollinearity among skills.

Even with these limitations, the results remain informative. Figure 6 illustrates the correlation coefficients between the top 50 skills and salary. Positive values indicate a stronger association with higher salaries, while negative values suggest a weaker or negative association. Skills like PyTorch, Machine Learning, TensorFlow, Kubernetes, Docker, and Python exhibit the highest positive correlations, underscoring their significant impact on boosting salary levels. This reflects the value given to expertise in AI, machine learning, and cloud technologies. Skills such as Active Directory, HTML, Excel, and Power BI show negative correlations with salary, suggesting that positions where these skills are required are typically linked to lower-paying job categories.

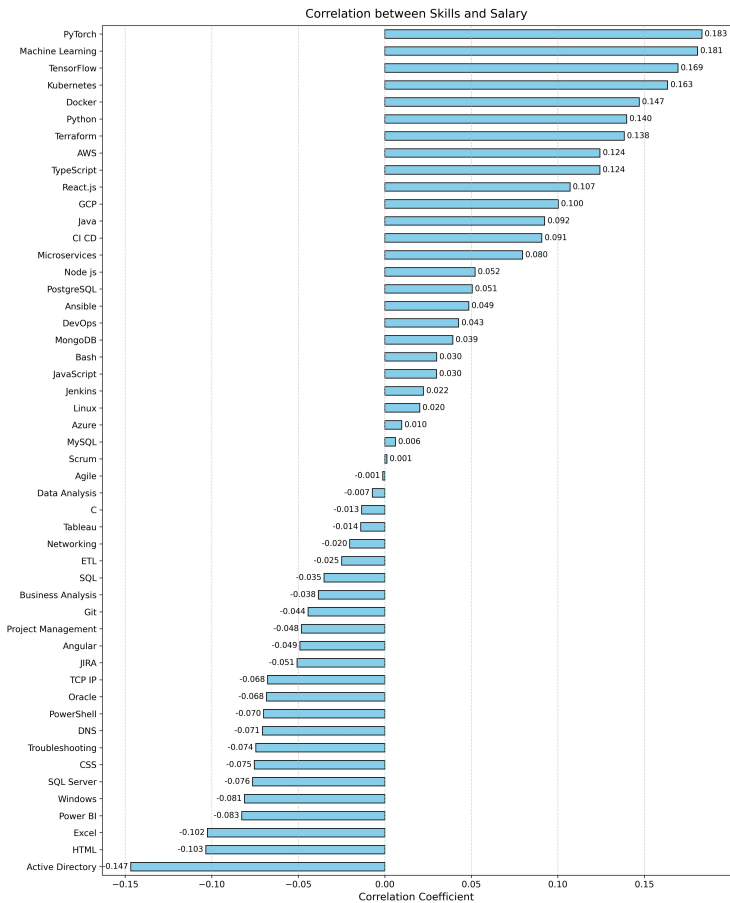


Figure 6. Correlation coefficient of Top 50 Skills x Salary.

Figure 7 presents a heatmap based on the phi coefficient (ϕ), a measure of association for binary variables. The phi coefficient is a special case of the Pearson correlation used with binary variables [29], indicating how frequently different technical skills are paired together in job postings. The coefficient reveals co-occurrence patterns, helping identify commonly paired tech industry skills. High positive correlations, such as those between

HTML and CSS, or TensorFlow, PyTorch, and Python, highlight commonly paired skills in web development and machine learning, respectively. In contrast, less related skills or belonging to different domains show weaker or negative correlations. For example, skills like Active Directory and HTML have minimal overlap with machine learning. There is a notable correlation between Java, C, and Python. Even though specialization in emerging technologies is highly valued, there remains significant space for professionals with expertise across multiple programming languages and technologies.

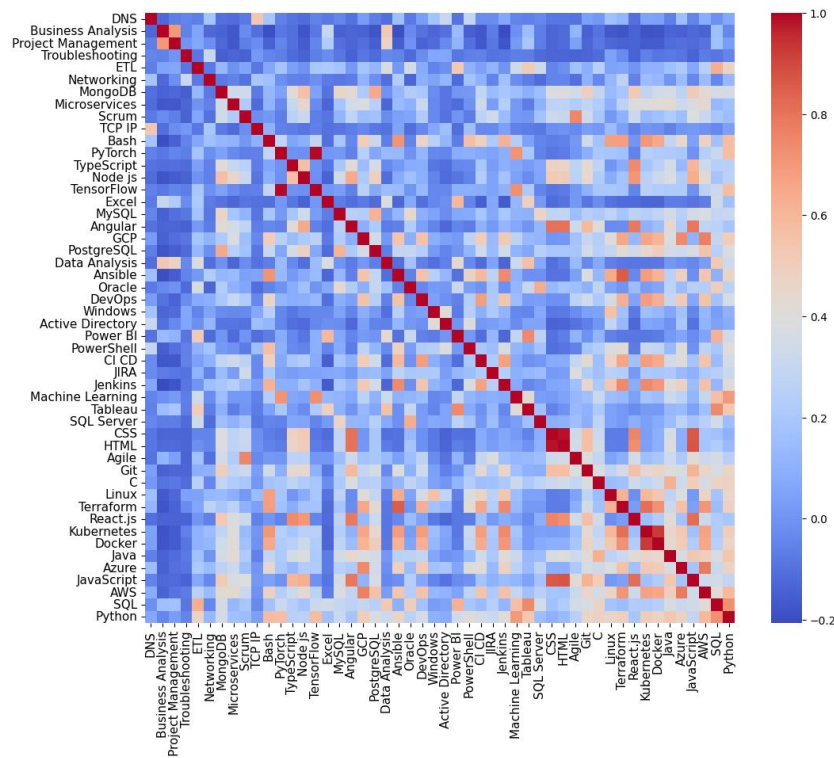


Figure 7. Heatmap of skill co-occurrence correlations.

To identify groupings of skills based on their co-occurrence in job postings, we applied K-Means clustering algorithm [30] and agglomerative clustering [31] to the co-occurrence matrix of the top 50 skills. The optimal number of clusters was determined by analyzing the inertia (within-cluster sum of squares) across different cluster counts. Based on the elbow method [32]—shown in Figure 8 (left), we selected five clusters, as presented in Table 2. The silhouette coefficients [33] for this configuration were very similar between methods (K-Means = 0.4001, Agglomerative = 0.4014), indicating that both algorithms produced clusters with comparable cohesion and separation.

Clusters SC 2 and SC 4 contain only one skill—Python and SQL, respectively. This outcome was anticipated, given that Figure 7 shows these skills as the most frequently mentioned, which likely contributed to this imbalance. In contrast, SC 3 includes a broad array of less frequent skills. SC 1 and SC 5 effectively highlight the grouping of complementary skills often found in job descriptions. The first cluster primarily comprises cloud and infrastructure-related tools, while the last cluster groups skills related to web development. The only discrepancy between the two clustering methods is the classification of Java: K-Means associates it with SC1 (cloud/infrastructure), whereas Agglomerative Clustering places it in SC5 (web development).

Table 2. Co-Occurrence Skill Clusters (K-Means baseline; differences under agglomerative clustering marked)

Cluster	Skills
SC 1	Git, Linux, Terraform, Kubernetes, Docker, Java [†] , Azure, AWS
SC 2	Python
SC 3	DNS, Business Analysis, Project Management, Troubleshooting, ETL, Networking, MongoDB, Microservices, Scrum, TCP/IP, Bash, PyTorch, TypeScript, Node.js, TensorFlow, Excel, MySQL, Angular, GCP, PostgreSQL, Data Analysis, Ansible, Oracle, DevOps, Windows, Active Directory, Power BI, PowerShell, CI/CD, JIRA, Jenkins, Machine Learning, Tableau, SQL Server, Agile, C
SC 4	SQL
SC 5	CSS, HTML, React.js, JavaScript

[†] Agglomerative clustering assigns *Java* to SC 5.

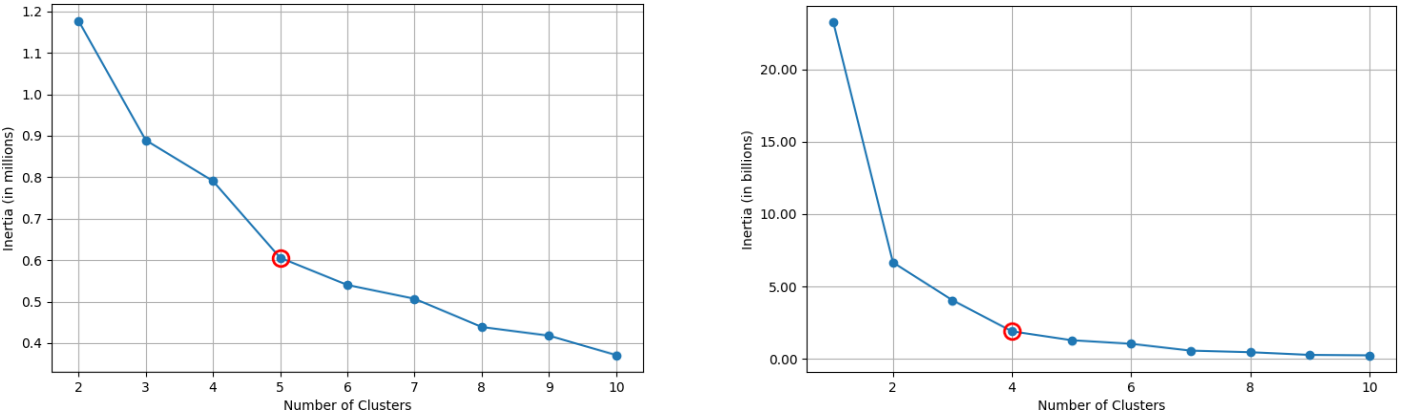


Figure 8. Inertia for co-occurrence skills clusters (left) and Salary-based clusters (right).

We also applied the clustering methods to group skills according to their associated median salaries, as shown in Table 3. Using the elbow method, illustrated in Figure 8 (right), four clusters were selected as the optimal number for this analysis. The Salary-based Skill Clustering (SSC) reveals a clear stratification aligned with market demand and compensation, as shown in Figure 9. The K-Means solution achieved a higher silhouette score (0.5731) compared to Agglomerative Clustering (0.5282), indicating that its clusters are more cohesive (skills inside each cluster are more similar to each other).

The highest-salary group (Cluster 3 in both methods) concentrates advanced and emerging technologies — such as Microservices, PyTorch, TensorFlow, Kubernetes, and Docker.. Below it, Cluster 1 groups cloud and DevOps skills such as MongoDB, GCP, PostgreSQL, Jenkins, and AWS, typically linked to platform engineering and automation roles. SSC 3 contains business and data-management skills, while SSC 6 aggregates support and networking skills (DNS, Excel, Active Directory), corresponding to the lowest salary tier.

Both methods produced identical results for Clusters 2 (SSC 3) and 4 (SSC 6). The only differences occur in Clusters 1 and 3: K-Means places some cloud/DevOps technologies (MongoDB, GCP, PostgreSQL) in the platform-engineering cluster (Cluster 1), whereas agglomerative shifts them into the highest-salary specialization cluster (Cluster 3).

Table 3. Salary-Based Skill Clusters

Cluster	Skills
SSC 1 — K-Means (Cluster 1)	MongoDB, Scrum, Bash, Node.js, GCP, PostgreSQL, Ansible, DevOps, CI/CD, Jenkins, Agile, Linux, React.js, Java, Azure, JavaScript, AWS, Python
SSC 2 — Agglomerative (Cluster 1)	Scrum, Bash, Node.js, Ansible, Jenkins, Agile, Linux, Java, Azure, JavaScript, Python
SSC 3 — Both Methods (Cluster 2)	Business Analysis, Project Management, ETL, Networking, MySQL, Angular, Data Analysis, Oracle, PowerShell, JIRA, Tableau, SQL Server, Git, C, SQL
SSC 4 — K-Means (Cluster 3)	Microservices, PyTorch, TypeScript, TensorFlow, Machine Learning, Terraform, Kubernetes, Docker
SSC 5 — Agglomerative (Cluster 3)	MongoDB, Microservices, PyTorch, TypeScript, TensorFlow, GCP, PostgreSQL, DevOps, CI/CD, Machine Learning, Terraform, React.js, Kubernetes, Docker, AWS
SSC 6 — Both Methods (Cluster 4)	DNS, Troubleshooting, TCP/IP, Excel, Windows, Active Directory, Power BI, CSS, HTML

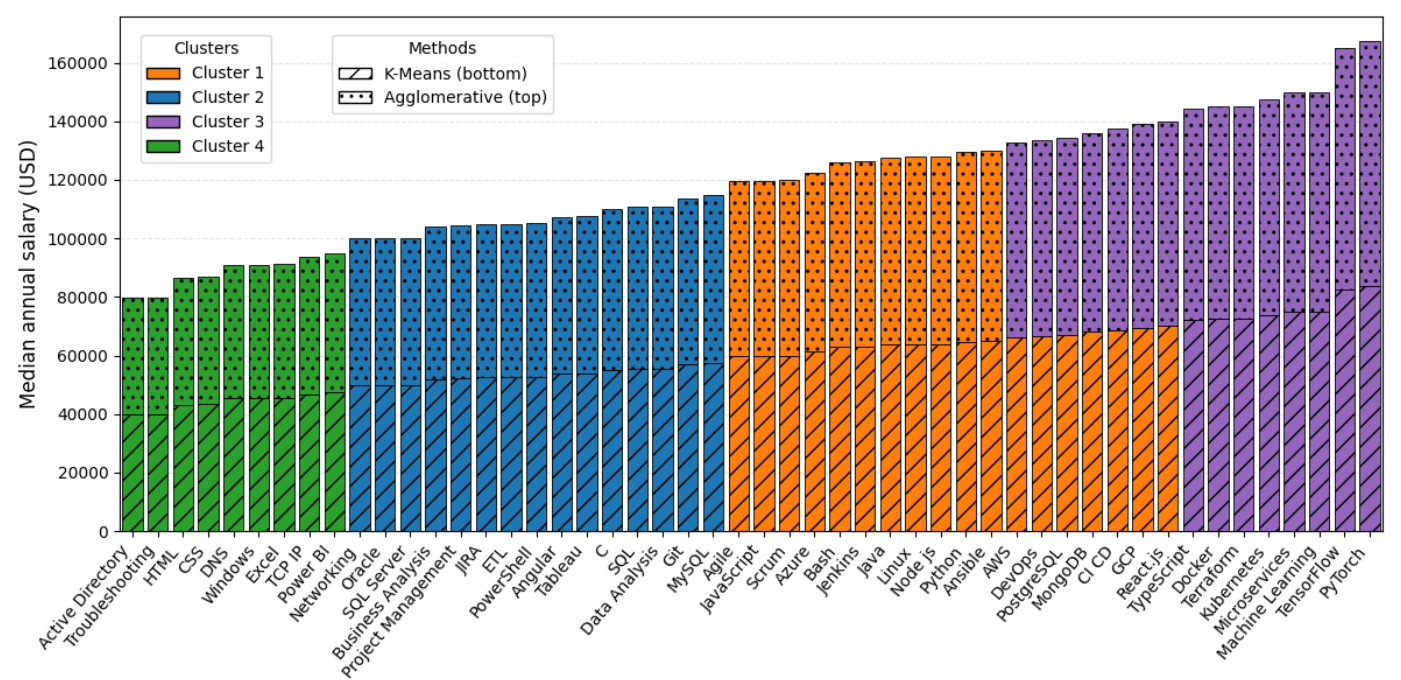


Figure 9. Median salary by skills in each cluster.

5. Discussion

While many of the findings, such as the prominence of Python, SQL, and AWS or the higher salaries for senior roles, may appear intuitive, the contribution of this study lies in quantifying these patterns with recent and large-scale evidence, and in uncovering less obvious nuances. The results demonstrate which skills dominate and the magnitude of their demand relative to others, the coexistence of traditional technologies with emerging ones, and the clear salary stratification associated with specific skills and roles. By mapping co-

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occurrence patterns, the analysis further highlights how employers combine competencies in practice, offering insight into complete job profiles rather than isolated requirements. These aspects add value by providing an updated, data-driven picture of the IT labor market that can help employers, professionals, and researchers understand current dynamics and plan for future developments.

As already noted in the methodology, a key limitation of this study is that all data were collected exclusively from LinkedIn. While LinkedIn offers global coverage and standardized search filters, its representativeness may vary across regions and industry sectors, as some markets rely more heavily on alternative recruitment platforms. Consequently, the findings presented here should be interpreted as reflecting patterns visible within LinkedIn rather than the entire IT labor market. Future research could broaden this scope by integrating data from other platforms to improve the robustness and generalizability of the results.

6. Conclusion

This work addresses the challenge of identifying trending skills in IT by analyzing job postings on LinkedIn. By leveraging large language models to extract and standardize skills and salary data, this work provides valuable insights for career planning and organizational training programs using an automated approach that also supports skill clustering.

Results indicate a significant variation in demand and compensation for different IT skills. The analysis revealed a right-skewed salary distribution, with most salaries clustering between \$50,000 and \$150,000. Notably, specialized skills in emerging technologies, such as machine learning and cloud computing, were associated with higher median salaries.

This work introduces an LLM-based approach for labor market analysis, specifically through prompt engineering, to extract and standardize unstructured data from LinkedIn job postings. This contribution directly addresses the challenge posed by LinkedIn's lack of structured information, enabling automated retrieval of technical skills and salary data at scale. From an academic standpoint, this strengthens the field of labor market analytics by offering a replicable and domain-agnostic framework, reinforcing its contribution to the intersection of knowledge management, artificial intelligence, and labor economics.

While this work focuses on IT job postings, the method demonstrates its capability to process heterogeneous and noisy textual descriptions, producing structured datasets suitable for statistical and trend analyses. The approach can be generalized beyond IT and extended to other domains, highlighting its potential as a flexible and domain-agnostic framework for skill and compensation mapping in dynamic labor markets. Therefore, from a practical standpoint, this work contributes to the understanding of current IT job market trends by providing a detailed analysis of the skills that employers and their corresponding salary ranges most seek after. The insights gained can assist individuals in making informed decisions about their career paths and skill development. Organizations can leverage this information to tailor their training programs to align with market demands, enhancing their competitive edge.

Nevertheless, several limitations must be acknowledged. The exclusive reliance on LinkedIn may introduce sampling bias across regions, sectors, seniority levels, and posting practices (e.g., optional salary disclosure), and the language of postings can reflect platform-specific communication patterns observed in prior LinkedIn-based studies [12,13]. The absence of ground-truth benchmarks or human adjudication limits the generalizability of the extracted skills and salaries. In addition, k-means is sensitive to initialization, feature scaling, and the choice of k , which may affect cluster stability and interpretation [30].

Future research should therefore pursue concrete methodological extensions. First, cross-platform collection and harmonization (e.g., integrating additional job boards) can

mitigate platform-specific bias and support comparability with broader online vacancy evidence [20]. Second, human–AI validation with expert-coded samples and inter-annotator agreement can quantify extraction accuracy and audit salary normalization steps. Third, robustness analyses of clustering should provide stability checks to assess sensitivity to modeling choices [29,32]. Finally, alignment with validated taxonomies can improve interpretability and portability across domains and regions [34,35]. Ultimately, this work acts as a methodological guide, establishing a flexible and domain-agnostic foundation that can be expanded to diverse labor markets, regions, and analytical tools for future research.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
API	Application Programming Interface
BI	Business Intelligence
CI/CD	Continuous Integration / Continuous Deployment
DNS	Domain Name System
ETL	Extract, Transform, Load
GCP	Google Cloud Platform
ICT	Information and Communication Technology
IT	Information Technology
KM	Knowledge Management
LLM	Large Language Model
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
OECD	Organisation for Economic Co-operation and Development
TCP/IP	Transmission Control Protocol / Internet Protocol

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